

ML- Experiment no 08 - Comparision of Classsification Algorithms

Import Libraries

```
In [2]: import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, precision_score, recall_score
```

```
In [3]: from sklearn.datasets import load_iris
```

```
In [7]: data = load_iris()
X = data.data
y = data.target
```

```
In [8]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2 , random_state = 42)
```

```
In [11]: scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.fit_transform(X_test)
```

```
In [15]: models = {
    "Logistic Regression": LogisticRegression(max_iter=200),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(random_state=42),
    "SVM": SVC(kernel='rbf', random_state=42),
    "KNN": KNeighborsClassifier(),
    "Naive Bayes": GaussianNB()
}
```

```
In [16]: results = []
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, average='macro')
    rec = recall_score(y_test, y_pred, average='macro')
    results.append([name, round(prec, 2), round(rec, 2), round(acc, 2)])
df_results = pd.DataFrame(results, columns=["Algorithm", "Precision", "Recall",
"Accuracy"])
print("\nComparison of Classification Algorithms:\n")
print(df_results)
```

Comparison of Classification Algorithms:

	Algorithm	Precision	Recall	Accuracy
0	Logistic Regression	0.97	0.97	0.97
1	Decision Tree	1.00	1.00	1.00
2	Random Forest	0.97	0.97	0.97
3	SVM	0.97	0.97	0.97
4	KNN	0.97	0.97	0.97
5	Naive Bayes	0.97	0.97	0.97

In []: